

Inflammatory Arthritis — Pattern-Based Approach

Place every inflammatory joint complaint on a 2-axis grid: TIME (acute vs chronic) × SPACE (mono = 1 · oligo = 2–4 · poly = 5+) — then let synovial fluid arbitrate.

The Time-and-Space Grid

	Mono / Oligo	Poly
Acute	Septic , gout, CPPD, Lyme (early), GC	GC, reactive, viral, sarcoid (Löfgren's)
Chronic	Fungal/TB, Lyme, tophaceous gout	RA, SLE, psoriatic, ankylosing spondylitis

Acute monoarthritis = septic until proven otherwise. Tap before steroids. Crystals do NOT exclude infection.

Synovial Fluid WBC — First Triage

WBC/μL	Think
<2K	Non-inflammatory (OA, derangement)
2–20K	Autoimmune, crystal, Lyme (>2K), fungal/TB (10–20K)
10–50K	Gonococcal (overlaps crystal)
>50K PMN	Septic (lower cutoff: prosthetic joint, prior abx)

- Send: cell count + diff, Gram stain, culture, crystal exam.
- **Crystals:** gout = needle-shaped, negatively birefringent MSU · CPPD = rhomboid, weakly positive CPP.

Acute — Can't-Miss & Crystal

Dx	Clues	Management
Septic	Hot swollen joint, ↓passive ROM, >50K PMN	GPC→vancomycin; GNR→ceftriaxone; urgent washout
Gout	1st MTP ~90%; obesity, HTN, CKD	Flare: NSAID/colchicine/steroid. ULT below.
CPPD	Chondrocalcinosis (knee, wrist); mimics gout/RA	Flares as per gout; screen PTH/Ca + iron studies
GC	Migratory arthralgia + tenosynovitis + pustules; 10–50K	Ceftriaxone (+ doxy if chlamydia not excluded, CDC 2021)
Reactive	1–6 wk post GI/GU infxn; HLA-B27; cultures neg	NSAIDs; eye/GU/skin ROS (keratoderma)
Viral	Parvo B19 IgM (RA-like); HBV/HCV; travel viruses	Supportive NSAIDs; serologies often misleading
Sarcoid	Löfgren's: E. nodosum + hilar LAD + arthritis	NSAIDs; MTX/TNFi if chronic

Gout ULT Rules (ACR 2020)

- **Start ULT:** ≥2 flares/yr, tophi, or erosions; consider after 1 flare if CKD ≥3, urate >9, or urolithiasis.
- **Allopurinol first-line** (incl. CKD) — start low, titrate. Goal urate <6 (<5 to shrink tophi).
- **Keep prophylaxis** (low-dose colchicine) while titrating — prevents mobilization flares.
- Tophaceous: punched-out erosions, overhanging edges; favors cool sites (fingertips, toes, Achilles, elbows).

Chronic — Infectious

Dx	Clues	Action
Fungal/TB	Indolent; immunosuppressed/ endemic travel; 10–20K, often culture-neg	Synovial biopsy; surgery + ID consult
Lyme	Knee mono/oligo, months post tick; fluid >2K	Borrelia PCR; doxycycline ×4 wk (IDSA 2020)

Chronic — Autoimmune Pattern Recognition

Dx	Pattern	Key clues / first-line
RA	Symmetric wrists/MCPs/ PIPs/MTPs; sparer DIPs + spine	RF, CCP, ESR/CRP; marginal erosions; MTX first-line
PsA	Asymmetric; DIPs + axial possible	Nail pits/onycholysis, dactylitis, enthesitis; skin ≠ joint severity
SLE	Symmetric small joints	Photosensitive rash, ulcers, alopecia, Raynaud's; swelling > erosions; HCQ
AS / axSpA	SI joints > spine; asymmetric peripheral	HLA-B27; syndesmophytes, ankylosis; TNFi (axial), SSZ (peripheral)

Screen the whole patient: rash, eye disease (uveitis), IBD/GI, GU symptoms, enthesitis, dactylitis — the extra-articular ROS is what separates the spondyloarthropathies.

Do / AVOID

- **DO** tap every acute monoarthritis before steroids; blood cultures if febrile.
- **DO** image: affected joint ± SI films if axial symptoms; hands/feet q1–2 yr in RA.
- **AVOID** ANA/RF to work up acute arthritis — viruses cause transient autoantibodies.
- **AVOID** ruling gout in/out by serum urate alone — may be normal during flares.
- **Red flags → urgent action:** fever + monoarthritis, prosthetic joint, IVDU, immunosuppression → tap + empiric abx + ortho; chronic >6 wk or erosions → rheumatology.